

Module-playerbots — Command Reference Manual

CLI Task Dispatcher & Operational Guide for Turtle WoW 1.18.1 / vMaNGOS PlayerBots

TARGET REPOSITORY

tortoise-wow-extended (mantech-turtle)

DISPATCHER SCRIPT

task.ps1 / tools/task.ps1

SUPPORTED GENERATOR

Visual Studio 17 2022 (x64 Release)

1. Execution Modes

All scanning, auditing, and synchronization workflows operate under three standard verification depths. When omitted, the execution mode **defaults to NORMAL**.

Mode	Inspection Scope	Selection & Filtration Criteria
FAST	Diff headers & metadata	Evaluates subject keywords and touched files. Selects high-confidence P0 crash and critical P1 fixes. Skips speculative features.
NORMAL (Default)	Full watermark range diffs & AST	Standard triage. Audits all commits since watermark. Selects P0 stability fixes, P1 AI/combat/healing parity, and mature P2 enhancements.
DEEP	Extended history, callers, blame	Deep semantic comparison across CMaNGOS and vMaNGOS. Traces caller chains, dependency trees, and checks for semantic equivalence.

2. Command Specifications

status

READ-ONLY

Displays total project health, target git branch, worktree cleanliness, upstream watermark locations, unscanned commit counts, and durable ledger statistics.

```
.\task.ps1 status
```

scan [cmangos | vmangos | all] [fast | normal | deep]

READ-ONLY

Audits newly unscanned upstream commits from the specified donor without modifying any target files. Automatically classifies candidate commits (P0-P3, CRITICAL, RECOMMENDED, EXPANSION_INCOMPATIBLE) and updates the durable ledger.

```
.\task.ps1 scan all normal
.\task.ps1 scan cmangos deep
.\task.ps1 scan vmangos fast
```

sync [cmangos | vmangos | all] [fast | normal | deep]

FULL PIPELINE

Executes the complete end-to-end synchronization pipeline: updates donor refs, audits candidate deltas in batch, then executes an atomic **commit-by-commit** loop (ports candidate natively, compiles `modules.lib`, links `mangosd.exe`, records in ledger, creates individual target commit, and immediately pushes each commit to `mantech-turtle`). Batch committing is strictly prohibited.

```
.\task.ps1 sync all normal
```

import-critical [cmangos | vmangos | all]

TARGETED PORT

Discovers, ports, and verifies only high-severity P0 fixes: server crashes, null pointer exceptions, memory corruption, and fatal bot lifecycle failures.

```
.\task.ps1 import-critical all
```

audit <source> <commit-sha>

READ-ONLY

Performs an isolated, deep semantic audit of one specific upstream commit. Evaluates target bug presence, Dungeon Clear impact, expansion compatibility, and adaptation steps.

```
.\task.ps1 audit cmangos 6ad783c9
.\task.ps1 audit vmangos ac4c83ad
```

port <source> <commit-sha>

TARGETED PORT

Directly ports an explicitly requested commit and required dependencies using target-native APIs, runs verification, and updates the ledger.

```
.\task.ps1 port cmangos 46cec841
```

verify-quick & verify-full

VERIFICATION GATE

Runs the target verification ladder:

- **verify-quick**: Rapid compilation of affected static module code (`modules.lib`).
- **verify-full**: Complete target server build and link of `mangosd.exe`.

```
.\task.ps1 verify-quick
.\task.ps1 verify-full
```

ledger [cmangos | vmangos]

REPORTING

Displays tabular status of all audited, ported, verified, duplicate, already-present, and rejected commits from state/porting-ledger.json.

```
.\task.ps1 ledger
.\task.ps1 ledger vmangos
```

roadmap

REPORTING

Displays active project milestones, phase progress (Phases 0-6), and prioritized backlog.

```
.\task.ps1 roadmap
```

3. Priority Tiers

Tier	Classification	Candidate Types
P0	CRITICAL	Crashes, null pointer dereferences, assertions, memory leaks, bot provisioning deadlocks.
P1	RECOMMENDED	Healer targeting priority, combat stances, threat calculations, pathing stalls, resurrect bugs.
P2	OPTIONAL / PENDING	Quality-of-life improvements, travel distribution, inventory/food adjustments.
P3	DISQUALIFIED	TBC/WotLK expansion mechanics, cosmetic noise, duplicate/already present code.

Verification Invariant: A source file compiling is never sufficient proof of success. mangosd.exe must link cleanly without unresolved externals before any commit is marked as verified or pushed to mantech-turtle.

Scope Boundary Invariant & Double-Confirmation Gate: Widening the scan scope into historical donor backlogs beyond the active watermarks is strictly **PROHIBITED**. If historical scope expansion is ever requested, the framework will not execute it immediately; it requires an explicit warning and a mandatory double-confirmation from the operator.

Mandatory Commit-by-Commit Pipeline (ADR-007): While candidate audits may be conducted in batches, all target code modifications, verification steps, git commits, and remote pushes must be executed strictly **commit-by-commit** (1 upstream donor commit = 1 atomic target commit = 1 remote push). Grouping multiple donor commits into a single commit or batch push is strictly prohibited.